

Complete Resource & Link Directory

Adaptive Variable-Resolution 2.5D LiDAR Mapping — every paper, dataset, doc, and library referenced, with direct links.

1. Research Papers

Paper	Venue/Year	Link
PointNet: Deep Learning on Point Sets for 3D Classification and Segmentation	CVPR 2017	https://arxiv.org/abs/1612.00593
PointNet++: Deep Hierarchical Feature Learning on Point Sets in a Metric Space	NeurIPS 2017	https://arxiv.org/abs/1706.02413
Cylinder3D: An Effective 3D Framework for Driving-scene LiDAR Semantic Segmentation	CVPR 2021 (Oral)	https://arxiv.org/abs/2008.01550
4D Spatio-Temporal ConvNets: Minkowski Convolutional Neural Networks	CVPR 2019	https://arxiv.org/abs/1904.08755
RangeNet++: Fast and Accurate LiDAR Semantic Segmentation	IROS 2019	https://www.ipb.uni-bonn.de/pdfs/milioto2019iros.pdf
SemanticKITTI: A Dataset for Semantic Scene Understanding of LiDAR Sequences	ICCV 2019	https://arxiv.org/abs/1904.01416
A Benchmark for LiDAR-based Panoptic Segmentation based on KITTI	ICRA 2021	https://arxiv.org/abs/2003.02371
Moving Object Segmentation in 3D LiDAR Data: A Learning-based Approach Exploiting Sequential Data	RA-L / IROS 2021	https://arxiv.org/abs/2105.08971
Receding Moving Object Segmentation in 3D LiDAR Data Using Sparse 4D Convolutions	2022	https://arxiv.org/abs/2206.04129
A Universal Grid Map Library: Implementation and Use Case for Rough Terrain Navigation (Fankhauser & Hutter — GridMap)	2016	https://www.research-collection.ethz.ch/handle/20.500.11850/188752
OctoMap: A Probabilistic, Flexible, and Compact 3D Mapping Framework Based on Octrees (Hornung et al.)	Autonomous Robots, 2013	https://octomap.github.io/octomap_paper.pdf
Building Variable Resolution Occupancy Grid Map from Stereoscopic System — A Quadtree Based Approach	—	https://www.researchgate.net/publication/261245976
Occupancy-Elevation Grid Mapping from Stereo Vision	—	https://www.researchgate.net/publication/261487271
Occupancy-elevation grid: an alternative approach for robotic mapping and navigation	Robotica, Cambridge Core	https://www.cambridge.org/core/journals/robotica/article/occupancyelevation-grid-an-alternative-approach-for-robotic-mapping-and-navigation/94FC3DD044922CE20CC074D9EADB18B9

Terrain-aware semantic mapping for cooperative subterranean exploration	Frontiers in Robotics and AI, 2023	https://www.frontiersin.org/journals/robotics-and-ai/articles/10.3389/frobt.2023.1249586/full
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2. Datasets

Dataset	Use	Link
SemanticKITTI PRIMARY	Point-wise semantic labels	https://semantic-kitti.org/
SemanticKITTI API	Loading, visualization, evaluation tools	https://github.com/PRBonn/semantic-kitti-api
nuScenes	3D boxes + temporal/tracking data	https://www.nuscenes.org/nuscenes
nuScenes devkit	Data loading/visualization utilities	https://github.com/nutonomy/nuscenes-devkit
KITTI (Vision Benchmark Suite)	Classical baselines, raw odometry	https://www.cvlibs.net/datasets/kitti/
Waymo Open Dataset	Large-scale evaluation (stretch goal)	https://waymo.com/open/

3. Core Libraries & Frameworks

Library	Purpose	Link
Open3D — Point Cloud Tutorial	I/O, filtering, transforms, visualization	https://www.open3d.org/docs/latest/tutorial/Basic/pointcloud.html
Open3D — GitHub repo	Source, examples, issues	https://github.com/isl-org/Open3D
Open3D — RANSAC plane segmentation docs	<code>segment_plane()</code> API reference	https://www.open3d.org/docs/latest/python_api/open3d.geometry.PointCloud.html#open3d.geometry.PointCloud.segment_plane
NumPy documentation	Vectorized array math	https://numpy.org/doc/stable/
PyTorch documentation	Tensors, autograd, DataLoader, GPU execution	https://docs.pytorch.org/docs/stable/
PyTorch Profiler recipe	Per-stage latency + GPU memory tracing	https://docs.pytorch.org/tutorials/recipes/recipes/profiler_recipe.html
scikit-learn — DBSCAN	Density-based clustering for obstacle instances	https://scikit-learn.org/stable/modules/generated/sklearn.cluster.DBSCAN.html
FilterPy	Kalman filter implementation (Python)	https://filterpy.readthedocs.io/en/latest/
FilterPy — GitHub repo	Source + companion book	https://github.com/rlabbe/filterpy
Kalman and Bayesian Filters in Python (companion book)	Conceptual grounding for tracking	https://github.com/rlabbe/Kalman-and-Bayesian-Filters-in-Python

4. Model Code & Pretrained Checkpoints

Model	What's here	Link
PointNet — official repo	Original TensorFlow code	https://github.com/charlesq34/pointnet

PointNet — PyTorch port	Common PyTorch reimplementation	https://github.com/fxia22/pointnet.pytorch
Cylinder3D — official repo	Code + pretrained checkpoints, SemanticKITTI/nuScenes	https://github.com/xinge008/Cylinder3D
Minkowski Engine — official repo	Sparse tensor library + install docs	https://github.com/NVIDIA/MinkowskiEngine
Minkowski Engine — documentation site	API reference	https://nvidia.github.io/MinkowskiEngine/
RangeNet++ / LiDAR-Bonnetal — training repo	Full training pipeline (PyTorch)	https://github.com/PRBonn/lidar-bonnetal
RangeNet++ — inference module (rangenet_lib)	C++/TensorRT inference wrapper	https://github.com/PRBonn/rangenet_lib

5. Multi-Resolution & Elevation Mapping Libraries

Library	Purpose	Link
GridMap (ANYbotics)	Multi-layer 2.5D elevation/traversability map library, ROS interface	https://github.com/ANYbotics/grid_map
GridMap — ROS 2 branch	ROS 2 (Humble/Iron/Rolling) compatible version	https://github.com/ANYbotics/grid_map/tree/ros2
Elevation Mapping (ANYbotics)	Robot-centric elevation mapping built on GridMap	https://github.com/ANYbotics/elevation_mapping
OctoMap	Multi-resolution octree occupancy mapping — library + docs	https://octomap.github.io/

6. Robotics & Visualization

Technology	Purpose	Link
ROS 2 documentation	Topics, tf2, bag playback	https://docs.ros.org/
RViz2 (via ROS 2 docs)	Point cloud / marker / grid visualization	https://docs.ros.org/en/rolling/Tutorials/Intermediate/RViz/RViz-User-Guide/RViz-User-Guide.html
Streamlit documentation	Python web dashboard framework	https://docs.streamlit.io/

7. Deployment & Optimization

Technology	Purpose	Link
NVIDIA TensorRT documentation	Model optimization/quantization for deployment	https://docs.nvidia.com/deeplearning/tensorrt/latest/

All links verified current as of this document's creation (Sept 2026). arXiv links resolve to the abstract page, which links to the PDF; GitHub links are to the default branch — check the README for branch-specific install instructions (e.g. GridMap's ROS 2 support lives on the `ros2/humble/iron/rolling` branches, not `master`).